NONEMPTY FINITE LIMITS FROM PAIR PRODUCTS, EQUALIZERS, AND FIBRE PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2250-2262, following the statement at 2239-2248.

Editorial note. The proof is complete; the equalizer identification is displayed data with a routine universal-property check recorded as an obligation.

Allowed standard prerequisites (editorial).

- Limits from products and equalizers. lemma-limits-products-equalizers, cited at proof lines 2253-2255. Supplies that existence of finite nonempty products and equalizers implies existence of finite nonempty limits.
- Nonempty finite limits include pair products, fibre products, equalizers. the limit definitions recalled at proof lines 2251-2252. Supplies the instantiation direction (1) implies (2) and (3).

Source reference. Original lemma 002N: sources/stacks-categories/pinned-original-source.tex, source lines 1745–1745. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category. The following are equivalent:*

(1) *Nonempty finite limits exist in $\mathcal{C}$.*
(2) *Products of pairs and equalizers exist in $\mathcal{C}$.*
(3) *Products of pairs and fibre products exist in $\mathcal{C}$.*

Proof. Since products of pairs, fibre products, and equalizers are limits with nonempty index categories we see that (1) implies both (2) and (3). Assume (2). Then finite nonempty products and equalizers exist. Hence by Lemma 002N we see that finite nonempty limits exist, i.e., (1) holds. Assume (3). If $a, b : A \to B$ are morphisms of $\mathcal{C}$, then the equalizer of a, b is

$$(A \times_{a,B,b} A) \times_{(\mathrm{pr}_1,\mathrm{pr}_2),A\times A,\Delta} A.$$

Thus (3) implies (2), and the lemma is proved. □

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.